

Unit 01: Introduction To Data Communication and Computer Networks

Rajni Bhalla, Lovely Professional University

CONTENTS

Objectives

Introduction

- 1.1 Data Communication
- 1.2 Defining Network
- 1.3 Benefits of computer network
- 1.4 Data Communication and Networking
- 1.5 Applications Of Computer Networks
- 1.6 Types of Networks
- 1.7 Difference Between LAN and WAN
- 1.8 Network Topologies

Summary

Keywords

Review Questions

Answers

Further/Suggested Readings

Objectives

After studying this unit, you will be able to:

- understand the basics of network and analyze the components required for communication.
- identify the different ways of representing the information and describe the main uses of computer networks
- learn different types of networks
- identify how one network differs from another network.
- explain the seven main topologies.

Introduction

The merging of computers and communications has a profound influence on the way systems are organized. The concept of a computer center as a room with a large computer to which the users bring their work for processing is now obsolete. The old model of a single computer servicing all the computational needs of an organization has been replaced by the one in which a large system of separate but interconnected computers does the job. These systems are called computer networks. The two computers are said to be interconnected if they can exchange information. The connection between the computers need not be only via a copper wire or fiber optics or microwaves. A communication satellite can be used for networking the computers.

History of Computer Networks

Following is a brief history of computers, networking, and telecommunication milestones:

1897: CRT (Cathode Ray Tube) credited to Braun

1900-1915: Teletype (telegraph 5 bit) 1915-1920: ARQ (Automatic Repeat request) credited to Van Duuren.

1915-1920: ARQ (Automatic Repeat request) credited to Van Duuren

1930-1940: ENIAC credited to DOD/MIT 1950s:

The 1950s: SAGE (Semi-Automatic Ground Environment) MIT 1950s

1960s: Transistorized Computers-2nd Generation 1961: CTSS (Compatible Time-Sharing System) credited to Cobato/MIT.

1965: Auto Equalization Techniques of Phone lines credited to Lucky et al. 1966: Fiber Glass credited to Kao & Hockman

1967: Integrated Circuits Computers-3rd Generation

1968: Carterfone-FCC Decision in

1969: A group of DoD researchers linked four computers at UCLA, SRI, University of Utah, and the UCSB. They created a network to communicate with one another about government projects. The network was part of the DoD's Advanced Research Project Agency and was dubbed ARPAnet

1972: More than 50 universities and military agencies were linked together on the network. For a short period, it was a top-secret defense project, ensuring that computers could talk to each other in the event of a nuclear attack. The communication system between the sites was called email and was invented by Ray Tomlinson of Bolt, Berank, and Newman.

1973: The defence project links were extended to Norway and England.

1974: Transmission Control Protocol (TCP) was published and the military and educational links diverged. Organizations like NASA began to experiment with computer networks, and the networks began to interconnect and the name Internet was coined.

1976: The Queen sends an email from RSRE Malvern.

1983: TCP/IP become the protocol standard for ARPAnet. Scott Fahlman invents the smiley to convey emotions in email.

1984: In the US, the NSF built high-speed, long-distance lines that connected supercomputer sites across the USA. These eventually replaced the original ARPAnet. In time, NSFnet was joined by other networks at dozens of universities, research laboratories, and high-tech companies. The system for assigning names to computers on the network was introduced – DNS. JANET was launched to connect British Universities.

1986: The NSF established its own faster network NSFnet and Network News Transfer Protocol (NNTP) was introduced making on-line interactive discussion a reality. The backbone speed was 56 Kbps.

1987: 1000 Notes th RFC and 10,000th host.

1988: Robert Tappan Morris releases the first Internet Worm and CERT was set up in response to this. Backbone speed upgraded to 1.544 Mbps. IRC developed.

1989: 100,000th host. Cuckoo's Egg was released by Cliff Stoll telling the true story of East German crackers accessing US installations.

1990: ARPAnet ceased to exist and the Internet effectively took its role.

1991: Gopher, a software program for retrieving information from servers on the Internet was made available by the University of Minnesota. The US Government announced that it no longer intended to restrict activity on the Internet to research. This policy shift was sufficient for 12 companies to co-operate and produce CIX. Phil Zimmerman released PGP. Backbone speed upgraded to 44.736 Mbps.

1992: The World Wide Web became a possibility after CERN, in Switzerland, released hypertext. 1,000,000th Host. The author gets his first dial-up email account with Demon Internet (Nov. 1992).

1993: Mosaic, a software program to browse Web sites written by Marc Andreessen, was released followed by Netscape.

1994: Shopping Malls arrive on the Internet. The UK Treasury goes online and the first cyberbank opens. The first banner adverts appeared for Zima (a drink) and AT&T.

1995: Traditional dialup services (AOL, CompuServe, etc) start to provide dialup services. The Vatican goes online. Several Internet companies go public. Netscape leads the field with the largest ever IPO on NASDAQ. DEC launches AltaVista, which claims to index every HTML page there is. Jeff Bezos launches Amazon.com. eBay is launched.

1996: 9,272 organizations find themselves unlisted after the InterNIC drops their name service as a result of not having paid their domain name fee. Various ISPs suffer extended service outages, bringing into question whether they will be able to handle the growing number of users. AOL (19 hours), Netcom (13 hours), AT&T WorldNet (28 hours - email only). China requires users of the Internet to register with the Police. Saudi Arabia restricts use to universities and hospitals. Domain name tv.com sold to CNET for US\$15,000. Backbone speed upgraded to 622 Mbps.

1997: 2000th RFC. 16 Million hosts. 1,000,000th Domain name registered (March 6th for Bonny View Cottage Furniture Company).

1998: 3,000,000th Domain name registered. US Postal authorities allow the purchase of postage stamps online for downloading and printing. Gigabit Ethernet standard ratified. Google is launched.

1999: First full-service bank opens on the Internet (First Internet Bank of Indiana). The first forged web page, looking like Bloomberg, raises the shares of a small company by 31% (7th April). Melissa strikes. 5,000,000th Domain name registered. First Cyberwar starts between Serbia and Kosovo. Shawn Fanning Launches Napster — record labels are furious.

2000: 10,000,000th Domain name registered. French Courts require that 'hate' memorabilia for sale on Yahoo's auction site must be removed. Gnutella is launched. ICANN selects new top-level domains. Backbone is upgraded to IPv6.

2001: Forwarding email becomes illegal in Australia (Digital Agenda Act). Napster was forced to suspend service after legal action. Taliban bans the Internet in Afghanistan. Nimda was released on the Internet.

2002: Distributed denial of Service attack hits 13 DNS root servers, causing national security concerns.

2003: The first official Swiss online election takes place in Anières (7 Jan), SQL Slammer (goes round the world in 10 minutes and takes out 3 of the 13 DNS Servers). Followed by SoBig.F (19 Aug) and Blaster (11 Aug).

2004: Lycos Europe releases a screen saver to help fight spam by keeping spam servers busy with requests (1 Dec). The service is discontinued within a few days after backbone providers block access to the download site and the service causes some servers to crash.

1.1 Data Communication

What is communication?

When two people communicate with one another that is known as communication as shown in Figure 1. When the sender is sending a message, They are sharing their ideas that are called communication.

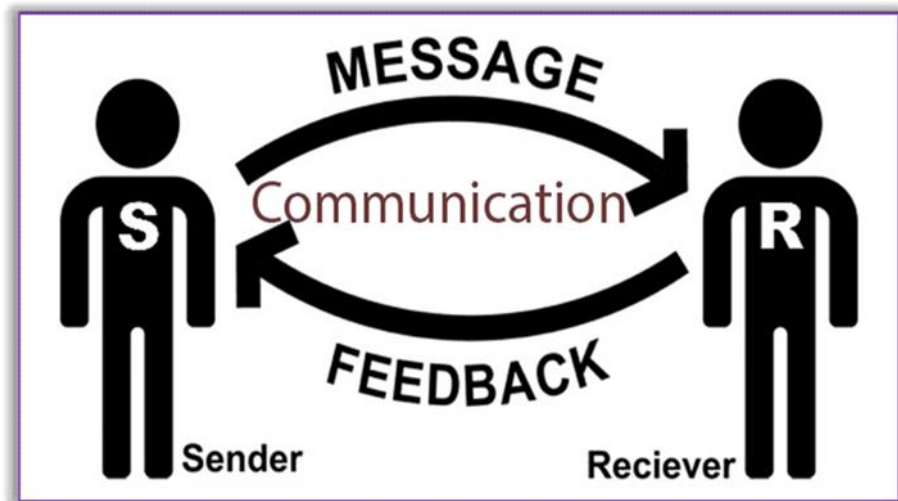


Figure 1 Data Communication

What is Data Communication?

When you are electronically transferring data from one place to another place by using a medium that is called data communication. When you are sharing data with another person that can be in any location or it can be in a remote location also. Communication always happens through some medium. If two people are communicating. We can say that medium is air. Similarly, when the sender wants to send data, the medium will be used to send data. That medium is called a transmission medium as shown in Figure 2. Medium is the physical path through which sender and receiver connect. A medium can be wired or wireless. Examples of the wired medium can be copper wire, twisted pair cable, and fiber optic cable as shown in Figure 3.

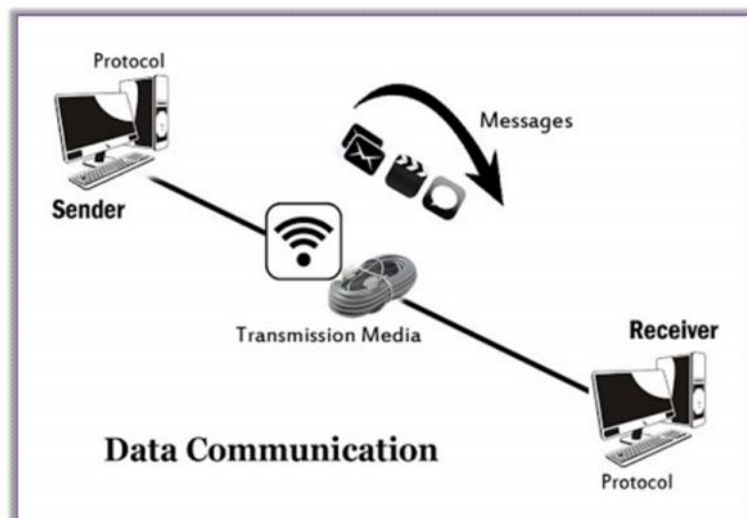


Figure 2 Transmission Media

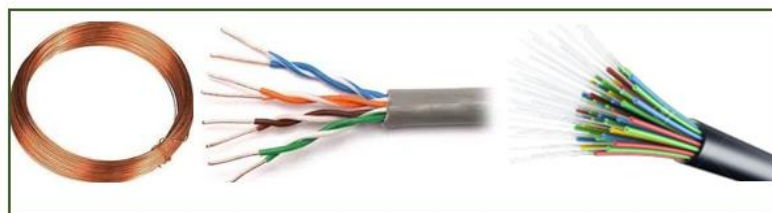


Figure 3 Medium for Data Transfer

1.2 Defining Network

A network consists of two or more computers that are linked to share resources (such as printers and CD-ROMs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams. The term 'computer network' means an interconnected collection of autonomous computers.

(a) Two computers are said to be interconnected if they can exchange information.

(b) The requirement for computers to be autonomous excludes from our definition systems in which there is a clear master/slave relation.

The key difference between a computer network and a distributed system:

In a distributed system, the existence of multiple autonomous computers is transparent to the user. A distributed system looks like a virtual uni-processor to its users.

With a network, a user must explicitly do the followings:

- ❖ log onto one machine (e.g., login),
- ❖ submit jobs remotely (e.g., rsh),
- ❖ move files around (e.g., rcp, FTP, UUCP),
- ❖ and generally handle all the network management personally.

In effect, a distributed system is a special case of a network, one whose software gives it a high degree of cohesiveness and transparency. The network is everywhere. The best examples of the network are railways, radio, water taps, and water supply chain systems in ancient Rome as shown in figure 5.



Figure 5 Best Examples of Network

What type of shared resources can be shared?

When one system sends data to another, it can share resources. Examples of resources are the internet, printer, and file server as shown in Figure 6.



Figure 6 Examples of Shared Resources

Shared resources can be in the form of hardware or they can in the form of software. Resources related to hardware are disks, printers, and scanners. Resources related to software are Files, applications that are also known, and application software as shown in Figure 7. Any type of file can be shared through the network. It can be in form of text, audio, and video.

Figure 4 Best examples of network

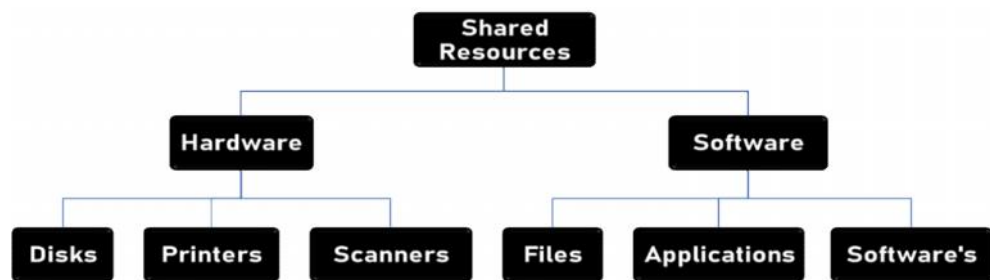


Figure 7 Shared Resources

For an example of hardware sharing, suppose if you only have one printer in your classroom that you want to share with several student computers, you can connect that printer to your computer and then share it so that it is accessible from any computer in the room. One of the major advantages is the utilization of the resource. Proper utilization of resources like the printer is doing here. No need to purchase a separate printer for each system. Maintenance costs also reduce when we share the resource.

1.3 Benefits of computer network

- File-Sharing: We can share a file with a different number of users.
- Hardware Sharing: If you are working in one network and if you share a file Best example is a printer. If 10 faculties are seated in a single room. Instead of purchasing an individual printer for each faculty, all can share a single printer. It reduces maintenance costs and increases resource utilization.
- Application Sharing: By using the internet, we can view and control the application that exists in a remote location. It means we can use applications that exist in some other system.
- User Communication: Two users can communicate with another. Two users can be connected using a router or switch or hub or any other device. They can be connected using peer-to-peer connection also.
- Network Gaming: Nowadays, it is very much important. All students joined in one location to play the game. They are located in different locations. Just to play a game, they are joining in a single location.

1.4 Data Communication and Networking

It is changing the way we do business. Day by day, business trending is changing. Today businesses are not limited to offline. When you want to extend your business, you want to opt online. The sellers are going to the online website so that they sell their products. Seller also wants to join their remote customer who is not in the same location. By this approach, users can expand their business. Due to competition in the market, business decisions have to be made more quickly. For this reason, decision-makers require immediate access to accurate information.

For example:

Why wait a week for that report from Germany to arrive by mail when it could appear instantaneously through computer networks. Business today relies on computer networks and internetwork.

Before we get into details of networking, we need to know: -

1. *How does the network operate?*
2. *What type of technologies are available?*
3. *Which design best fills which set of needs?*

It means what type of network you wanted to build. Suppose if you wanted to connect the system to a small institute. You need to decide either you are going to connect all systems either using a bus topology, ring topology or you want to connect using a hybrid topology. How you are going to connect all systems. To make connectivity between all systems together, basic knowledge is required. If you want all systems able to shared resources and data with one another, it means technology knowledge is very much required. You must know what type of technology is required, which is the best technology to connect systems.

The development of the personal computer brought tremendous changes for:

- Business
- Industry
- Science
- Education

It has changed the way people are looking. We are looking for online education. Students are looking for specialized certificate courses. Similarly, revolution is occurring in data communication and networking. Technology advances are making it possible for communications links to carry more and faster signals. Services are evolving to allow the use of this expanded capacity.

Example: Telephone services extended to have:

- Conference calling
- Call waiting
- Voice mail and
- Caller ID

4. *What is the role they are playing in the research?*

Research in data communications and networking has resulted in new technologies. One goal is to be able to exchange data such as text, audio, and video from all points in the world.

5. *Why would we want to access the internet?*

We want to access the internet to download and upload information quickly and accurately and at any time.

6. *What is communication*

When two people connected through a medium so that they can communicate and share knowledge. Sharing can be local or remote.

Local Sharing

Local sharing means two users are directly communication with one another. Example face to face communication as shown in Figure 8.

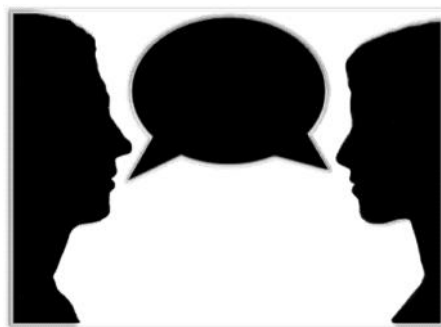


Figure 8 Local Communication

Remote Sharing

Remote sharing means when we talk about distance. User with whom we want to communicate that is not located locally. The user is located at a remote location. It means the user is located at some other place. Example telecommunication means telephony, telegraphy, and television means communication at a distance as shown in Figure 9.



Figure 9 Remote Sharing

What is data?

Data refers to information presented in any form and is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as wire cable. For data communication to occur the communicating devices must be a part of a communicating system made up of a combination of hardware (physical equipment) and software (programs).

The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

- **Delivery** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user. If it's not reaching to correct destination then security is compromised here.
- **Accuracy:** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable. When the sender is sending data, in between no manipulation of data is allowed.
- **Timeliness:** The system must deliver data promptly. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called **real-time transmission**. Especially in case if you are sending audio and video, data must deliver in time. For example, if we play a video and audio is coming after few minutes. So, I cannot enjoy listening to that video. There should not be much delay of data.
- **Jitter:** Jitter refers to the variation in the packet arrival time. It is an uneven delay in the delivery of audio or video packets. There should not be many variations in the early packet and coming packet as shown in Figure 10. Variations in the packet should less. Jitter in between first and second packet is of 5 minutes and second to third is also 5 minutes and third, to the fourth packet is taking 15 minutes. It is not at all acceptable. Jitter means variation in the packet delay should be consistent as shown in Figure 10.

Components of Data Communication

Five components of data communication play a major role as shown in figure 11:

- Sender
- Receiver
- Message
- Transmission medium
- Protocol

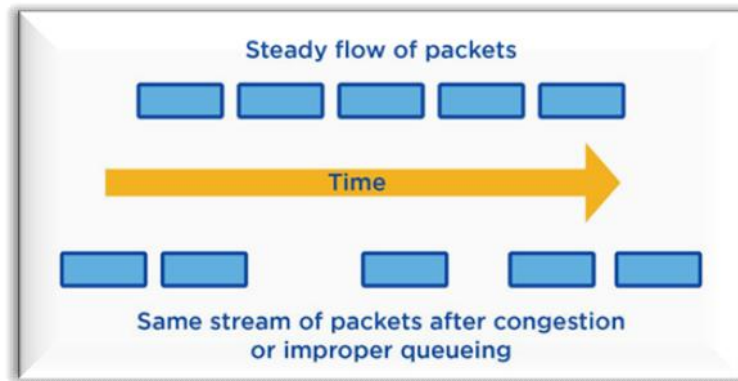


Figure 10 Jitter Variation in Packets

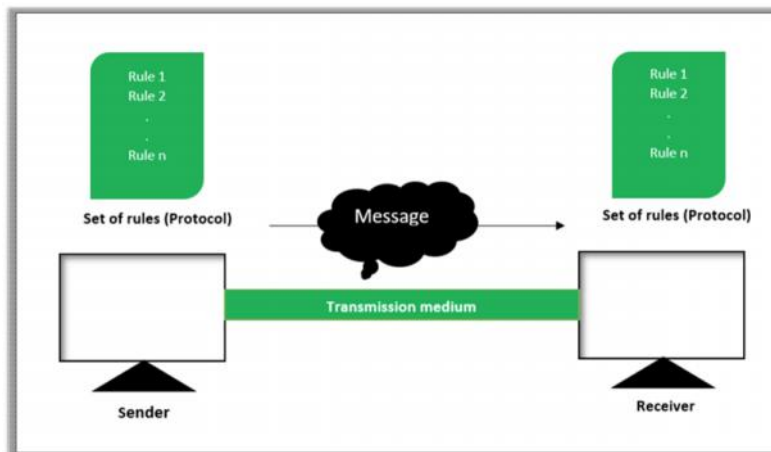


Figure 11 Components of Data Communication

Sender: Sender who sends the data. It can be a workstation from where you are sending the data.

Receiver: Receiver who has the capability of receiving the data. It can also be a workstation that is receiving the data.

Message: The information that users want to send and that can be in the form of text, audio, and video.

Transmission medium: it can be either wired or wireless medium. Wired media are also known as guided media and wireless media means without wire. Examples of wireless media can be microwave, radio wave, Bluetooth, or mobile communication.

Protocol: The sender is going to use some rules and regulations to send a message to the destination. Communication should happen by following some rules. For example: If the sender knows only French and the receiver knows only English. They have to follow some rules to communicate with one another. Otherwise, they won't be able to communicate with one another.

Data Representation

Information can be represented in text, images, numbers, audio, and videos form.

Text is a collection of alphabets. Smaller case and uppercase alphabets are also allowed. These alphabets will be converted into the form of bits. Information can be represented in the form of images also that would be converted into bits. Numbers are in the form of digits. Information can be represented in the form of audio. If you wanted to send recorded sound that can also be sent. When you want to send a video to your friend that can also be in the form of bits.

1.5 Applications Of Computer Networks

The main uses of computer networks are in business, home, mobile and social issues.

Business Applications

Most of the companies have a substantial number of computers. There uses are

- **Resource Sharing:**

The first one is resource sharing that helps to share the resource. Computer networks allow organizations to have units that are placed apart from each other, to share information in a very effective manner. Programs and software in any computer can be accessed by other computers linked to the network. It also allows sharing of hardware equipment, like printers and scanners among varied users as shown in Figure 12.

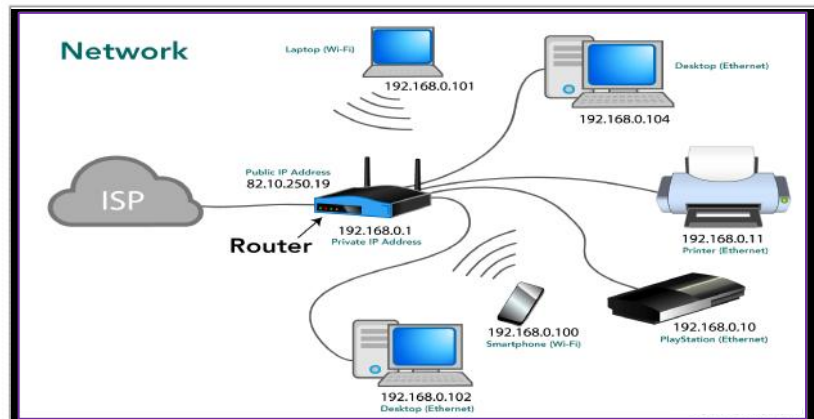


Figure 12 Resource Sharing

- **Server Client Model:**

A client-server model is shown in Figure 13 which consists of two clients and one server and they are connected by a network.

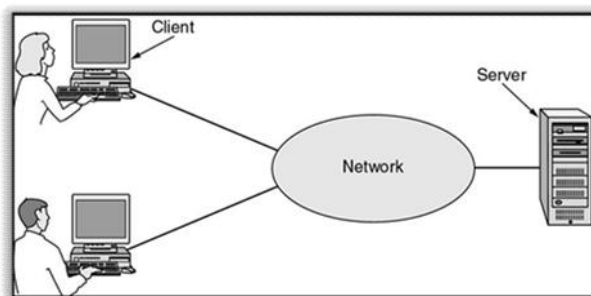


Figure 13 Client/Server Model

The client is sending a request and the server is replying because they are connected through a network. The client-server model describes how a server provides resources and services to one or more clients. Examples of servers include web servers, mail servers, and file servers. Each of these servers provides resources to client devices, such as desktop computers, laptops, tablets, and smartphones.

- **Communication medium**

A computer network can provide a powerful communication medium among employees. Virtually every company that has two or more computers now has e-mail (electronic mail), which employees generally use for a great deal of daily communication as shown in.



Figure 14 Communication Medium

- E-commerce**

A goal that is starting to become more important in businesses is doing business with consumers over the Internet.



Figure 15 Examples of E-commerce

Airlines, bookstores, and music vendors have discovered that many customers like the convenience of shopping from home as shown in Figure 15. This sector is expected to grow quickly in the future. The most popular forms in E-commerce are listed in

Table 1 Most Popular Forms in E-commerce

The most popular forms are listed

B2C (Business to Consumer)	When consumer order shoes online
B2B (Business to Business)	Truck manufacturer ordering tires from suppliers
C2C (Consumer to Consumer)	An online auction site (ebay)
G2C (government to Citizen)	reduce the average time for fulfilling citizen's requests for various government services.
P2P (Peer to Peer)	Buyer and seller transact directly

- P2P Communication**

Stands for "Peer to Peer." In a P2P network, the "peers" are computer systems that are connected via the Internet. Files can be shared directly between systems on the network without the need for a central server. In other words, each computer on a P2P network becomes a file server as well as a client.

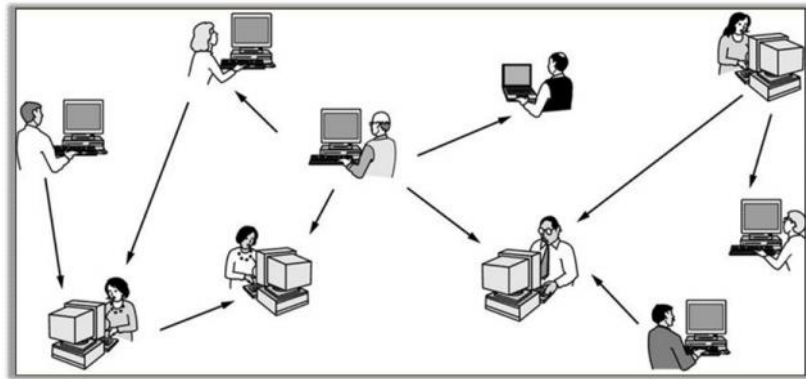


Figure 16 Peer to Peer Communication

Once connected to the network, P2P software allows you to search for files on other people's computers. Meanwhile, other users on the network can search for files on your computer, but typically only within a single folder that you have designated to share. While P2P networking makes file sharing easy and convenient, it also has led to a lot of software piracy and illegal music downloads. Therefore, it is best to be on the safe side and only download software and music from legitimate websites.

Home Applications

Some of the most important uses of the Internet for home users are as follows:

- *Access to remote information:*
- *Person-to-person communication*
- *Interactive entertainment*
- *Electronic commerce*

Mobile Users

Mobile computers, notebook computers are one of the fastest-growing segments of the entire computer industry. Since having a wired connection is impossible in cars, boats, airplanes, there is a lot of interest in wireless networks. Wired connection is not possible in these areas, so wireless connection comes into the picture. For example, people who usually traveling, they prefer to use portable devices.



Figure 17 Wireless Hotspots

Wireless hotspots are another kind of wireless network for mobile computers Figure 17. Smartphones such as the popular iPhone, combine aspects of mobile phones and mobile computers. Although wireless networking and mobile computing are often related, they are not identical, as the below Figure 18 shows:

Wireless	Mobile	Applications
No	No	Desktop computer in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Figure 18 Mobile Users

Social Issues

The growth in the availability of affordable computing technology has caused several major shifts in the way that society operates. The majority of these have been for the better, with home computers and the internet providing unlimited access to all of the information ever created and discovered by humanity.

- **Communication Breakdown**

Socializing within a family unit has always been important, as it strengthens the bonds between us and ensures cohesion within the group. But with more and more households owning several computers and numerous portable devices granting access to information and entertainment, some argue that this is leading to a lack of family communication. If each member is engrossed in their laptop, smartphone, or tablet each evening, even communal things like watching television are compromised. Meanwhile, you can see whole families who are out to dinner and still staring into a touchscreen rather than talking to one another. And if you're the one driving to that family dinner and texting while driving, you're a distracted driver, increasing your risk of crashing, and potentially causing death and injury. Increase your digital wellbeing by allowing technology to improve your life and not become a distraction to your life and others. Your life and others are more important than technology.

- **Unauthorized Access**

Unauthorized access is when someone gains access to a website, program, server, service, or other system using someone else's account or other methods. For example, if someone kept guessing a password or username for an account that was not theirs until they gained access, it is considered unauthorized access.

- **Authentication**

Authentication is the process of recognizing a user's identity. It is the mechanism of associating an incoming request with a set of identifying credentials. The credentials provided are compared to those on a file in a database of the authorized user's information on a local operating system or within an authentication server.

- **Ethical Use of Networks**

It is a generally accepted policy to maintain access to the **networks** as open as possible; it is thus unavoidable that certain **ethical** standards be imposed on the activities of individual users. Such demands differ little from those placed on other members of a modern civilized community.

- **Identity Theft**

Identity theft, also known as *identity fraud*, is a crime in which an imposter obtains key pieces of personally identifiable information (PII), such as Social Security or driver's license numbers, to impersonate someone else. The taken information can be used to run up debt purchasing credit, goods, and services in the name of the victim, or to provide the thief with false credentials. In rare

cases, an imposter might provide false identification to police, creating a criminal record or leaving outstanding arrest warrants for the person whose identity has been stolen.

- **Cyberbullying**

Cyberbullying or cyberharassment is a form of bullying or harassment using electronic means. Cyberbullying and cyberharassment are also known as online bullying. It has become increasingly common, especially among teenagers, as the digital sphere has expanded and technology has advanced.

- **Gaming Addiction**

Video game addiction is the compulsive or uncontrolled use of video games, in a way that causes problems in other areas of the person's life. Often considered a form of a computer addiction or internet addiction, video game addiction has been an increasing concern for parents as video games have become more commonplace and are often targeted at children.

- **Health & Fitness**

Fitness involves the activity of some sort that stimulates various systems of the body and maintains a certain condition within the body. Health, on the other hand, involves every system of the body and is only achieved through a lifestyle that supports health.

1.6 Types of Networks

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Wide Area Network (WAN)
- Metropolitan Area Network (MAN)

Personal Area Network (PAN)

A personal area network (PAN) is the interconnection of information technology devices within the range of a person, typically within a range of 10 meters. For example, a person traveling with a laptop, a personal digital assistant (PDA), and a portable printer could interconnect them without having to plug anything in, using some form of wireless technology. Typically, this kind of personal area network could also be interconnected without wires to the Internet or other networks as shown in Figure 19.

PAN is organized around a person within a single building. This could be inside a small office or residence. PAN would include one or more computers, telephones, peripheral devices, video game consoles, and other personal entertainment devices. Two or more devices are located on your person and normally connect via Bluetooth.

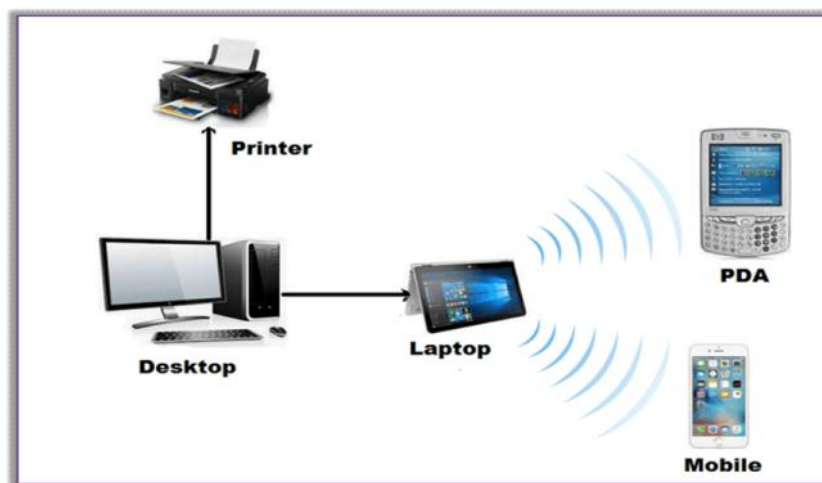


Figure 19 Personal Area Network

Types of PAN

PAN can be of different types. It can be wired PAN or Wireless PAN defined in diagram below:

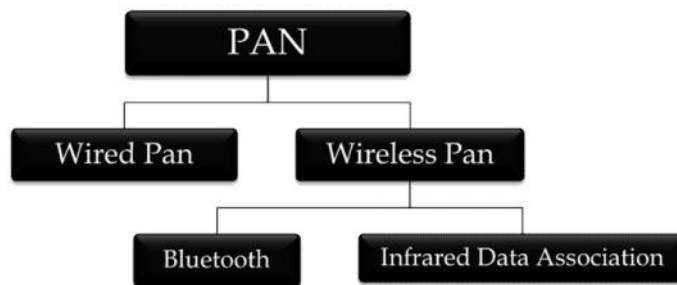


Figure 20 Types of PAN

Wired PAN

- The data cable is an example of the above PAN.
- This is also a Personal Area Network because that connection is for the user's personal use. PAN is used for personal use only.

Wireless Personal Area Network

The two kinds of wireless technologies used for WPAN are Bluetooth and Infrared Data Association. **Bluetooth** uses short-range radio waves over distances up to approximately 10 meters. For example, Bluetooth devices such as a keyboard, pointing devices, audio headsets, printers may connect to personal digital assistants (PDAs), cell phones, or computers wirelessly.



Figure 21 Examples of Bluetooth and Infrared data association

Infrared Data Association (IrDA) uses infrared light, which has a frequency below the human eye's sensitivity. Infrared, in general, is used, for instance, in TV remotes.

Local Area Network

LAN is normally located inside the same building or structure or residence, school, laboratory, or office building. A LAN can be as simple as two computers connected to a single switch. A LAN is very useful for sharing resources, such as data storage and printers. LANs can be built with relatively inexpensive hardware, such as hubs, network adapters, and Ethernet cables. High speed and relatively low cost are the defining characteristics of LANs. It is usually privately owned networks. LANs are typically used for single sites where people need to share resources among themselves but not with the rest of the outside world as shown in Figure 22.

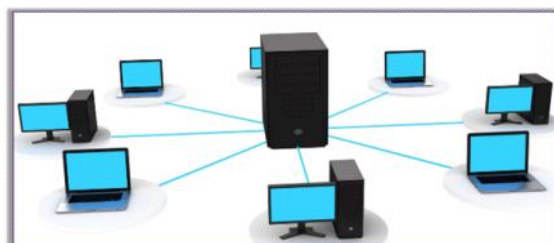


Figure 22 Local Area Network

Two most common transmission technologies in use for local area networks:

- Ethernet over twisted-pair cabling

- Wi-Fi



Historical technologies include ARCNET, Token Ring, and AppleTalk as shown in Figure 23.

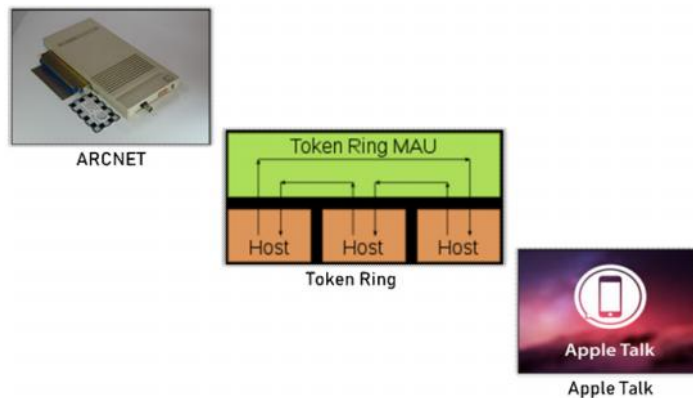


Figure 23 Historical Technologies of LAN

RELATED TO LAN IS WLAN

Another name for WLAN is wi-fi(Wireless Fidelity).

Classification of LAN

Computer networks can logically be classified into 1) peer-to-peer networks and 2) client-server networks as shown in Figure 24.

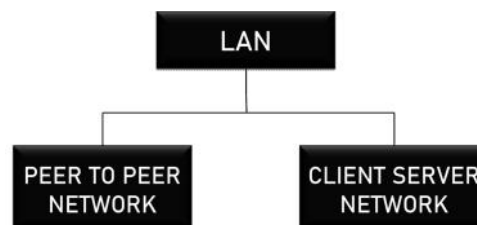


Figure 24 Types of LAN

Peer to Peer Network

Each computer is responsible for making its resources available to other computers on the network. Responsible for setting and maintaining own security for these resources. Each computer is responsible for accessing the required resources from peer-to-peer relationships. Peer to peer network is usually for a small network containing less than 10 computers on a single LAN. Do not have a central control system. There are no servers in peer networks as shown in Figure 25. Peer to Peer networks is amplified into homegroups.

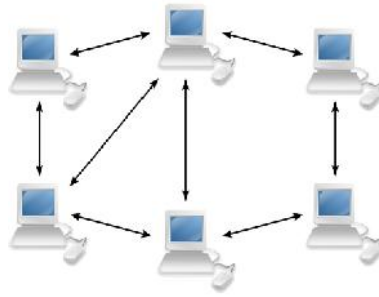


Figure 25 Peer to Peer Network

Advantages and disadvantages of a peer network

Table 2 Advantages and Disadvantages of Peer Network

Advantages	Disadvantages
Use less expensive computer hardware	Not Very Secure
Easy to administer	No central point of storage or file archiving
No NOS required	The additional load on the computer because of resource sharing
More built-in redundancy, easy setup, and low cost	Hard to maintain version control.

Client/Server Network

A computer network in which one centralized, powerful computer (called the server) is a hub to which many less powerful personal computers or workstations (called clients) are connected. The clients run programs and access data that are stored on the server.

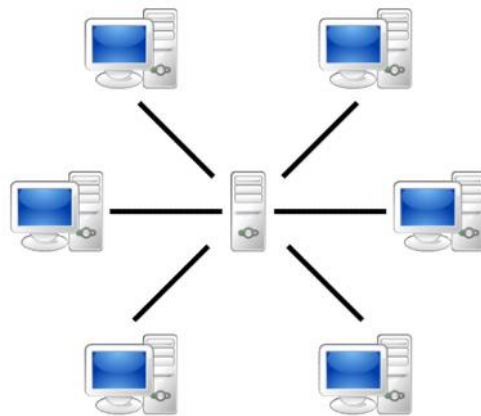


Figure 26 Client/Server Network

Table 3 Advantages and Disadvantages of Client/Server

Advantages	Disadvantages
Very secure	Requires professional administration
Better performance	More hardware intensive
Centralized backup	More software-intensive

Very reliable

Expensive dedicated software

Metropolitan Area Network

A **metropolitan area network (MAN)** is a computer network larger than a local area network. MAN knew as *municipal area network*. consists of a computer network across an entire city, college campus, or small region. A MAN is often used to connect several LANs to form a bigger network. A MAN is larger than a LAN, which is typically limited to a single building or site. An example of MAN is shown in Figure 27 below.

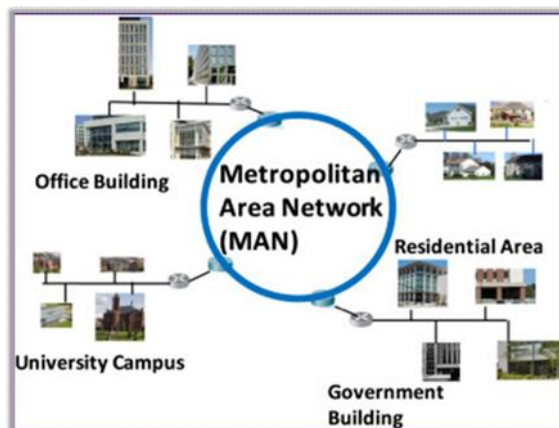


Figure 27 Metropolitan Area Network

Networking technologies used in municipal networks

- Asynchronous Transfer Mode (ATM),
- FDDI,
- and SMDS(Switched Multi-megabit Data Service)

Wide Area Network

A **wide area network**, or **WAN**, occupies a very large area, such as an entire country or the entire world. A WAN can contain multiple smaller networks, such as LANs or MANs. The Internet is the best-known example of a public WAN.

How WAN networks are established?

Computers connected to a wide-area network are often connected through public networks, such as the telephone system. They can also be connected through leased lines or satellites. The largest WAN in existence is the Internet.

1.7 Difference Between LAN and WAN

Table 4 Difference Between LAN and WAN

LAN	WAN
Owned by a person=Privately owned	Can be private or public
Operate over small area	Large distance across countries.
Easy to design and maintain	Not Easy
Communication medium is generally coaxiable cables.	Satellite links(because of countries)
Minimum propagation delay(time required for a digital signal to travel from the input(s) of a logic gate to the output)	Excessive propagation delay

High data rate(because of small distance)	Low data rate
Broadcasting	Switching techniques

Table 5 Difference Between LAN, MAN, and WAN

Parameters	LAN	WAN	MAN
Ownership of network	Private	Private or Public	Private or Public
Geographical area covered	Small	Very large	Moderate
Design and maintenance	Easy	Not easy	Not easy
Communication medium	Coaxial cable	PSTN or satellite links	Coaxiable cables,PSTN,optical fiber cable,wireless
Bandwidth	Low	High	Moderate
Data Speed	High	Low	Moderate

1.8 Network Topologies

A Network Topology is the arrangement with which computer systems or network devices are connected. Topologies may define both physical and logical aspects of the network. Both logical and physical topologies could be the same or different in the same network.

- *The physical topology* is the placement of the various components of a network, including device location and cable installation,
- while *logical topology* illustrates how data flows within a network, regardless of its physical design

Point-to-Point

The simplest topology with a dedicated link between two endpoints. The simplest topology is a permanent link between two endpoints. A children's tin can telephone is **one example** of a physical dedicated channel.

Types of Topologies

There are seven main topologies as shown in Figure 28 below:

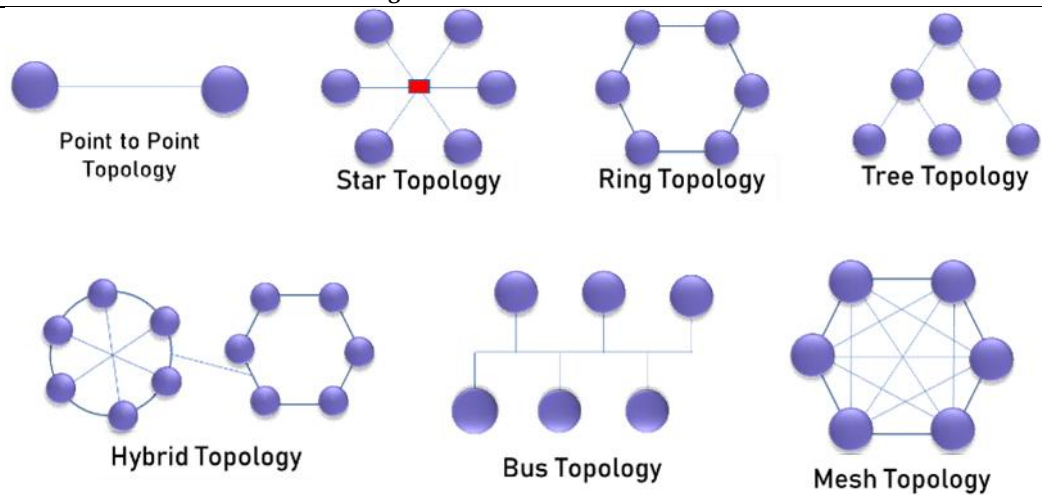
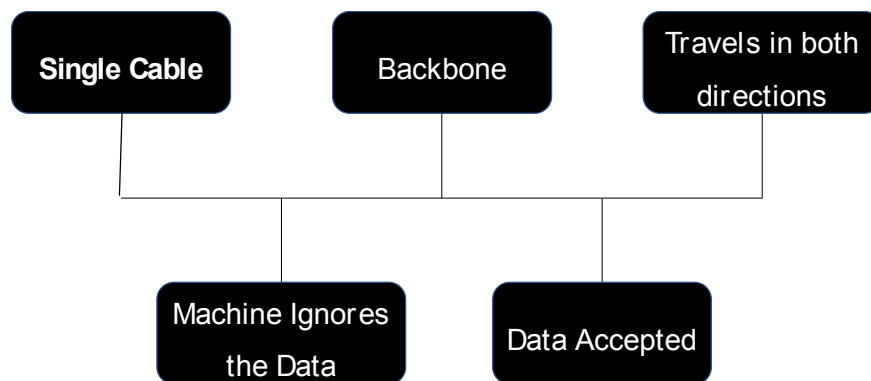


Figure 28 Types of Topologies

Bus Topology

In a bus topology, each node is connected to a single cable, with the help of interface connectors. This central cable is the backbone of the network and is known as the bus. A signal from the source travels in both directions to all machines connected on the bus cable until it finds the intended recipient. If the machine address does not match the intended address for the data, the machine ignores the data. Alternatively, if the data matches the machine address, the data is accepted.



The benefit over other technology

Because the bus topology consists of only one wire, it is rather inexpensive to implement when compared to other topologies. Because only one cable is utilized, it can be the single point of failure.

Working of Bus Topology

A signal from the source is broadcasted and it travels to all workstations connected to bus cable. Although the message is broadcasted only the intended recipient, whose MAC address or IP address matches, accepts it. If the MAC /IP address of the machine doesn't match with the intended address, the machine discards the signal. A terminator is added at the ends of the central cable, to prevent the bouncing of signals.

Advantages (benefits) of Linear Bus Topology

- 1) It is easy to set-up and extends the bus network.
- 2) The cable length required for this topology is the least compared to other networks.
- 3) Bus topology costs very little.
- 4) Linear Bus network is mostly used in small networks. Good for LAN.

Disadvantages (Drawbacks) of Linear Bus Topology

1. There is a limit on central cable length and the number of nodes that can be connected.
2. Dependency on central cable in this topology has its disadvantages. If the main cable (i.e. bus) encounters some problem, the whole network breaks down.
3. Proper termination is required to dump signals. The use of terminators is a must.
4. It is difficult to detect and troubleshoot faults at the individual station.
5. Maintenance costs can get higher with time.
6. The efficiency of Bus network reduces, as the number of devices connected to it increases
7. It is not suitable for networks with heavy traffic.
8. Security is very low because all the computers receive the sent signal from the source.

Star Topology

In local area networks with a star topology, each network host is connected to a central hub with a point-to-point connection. So it can be said that every computer is indirectly connected to every other node with the help of the hub. In Star topology, every node (computer workstation or any other peripheral) is connected to a central node called a hub, router, or switch. The switch is the server and the peripherals are the clients. The network does not necessarily have to resemble a star to be classified as a star network, but all of the nodes on the network must be connected to one central device as shown in Figure 29. All traffic that traverses the network passes through the central hub. The hub acts as a signal repeater. The star topology is considered the easiest topology to design and implement. Devices typically connect to the hub with Un-shielded Twisted Pair (UTP) Ethernet.

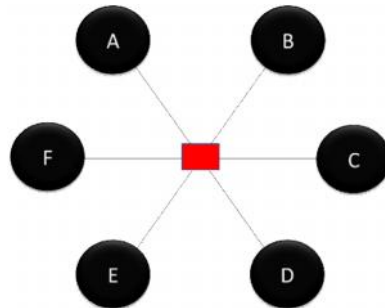


Figure 29 Star Topology

Primary Disadvantage Compared to Other

The primary disadvantage of the star topology is that

1. The hub represents a single point of failure.
2. Compared to the bus topology, a star network generally requires more cable.
3. The use of a hub, a router, or a switch as a central device increases the overall cost of the network.
4. The use of a hub, a router, or a switch as a central device increases the overall cost of the network.

Advantages of Star Topology

- As compared to Bus topology it gives far much better performance, signals don't necessarily get transmitted to all the workstations. A sent signal reaches the intended destination after passing through no more than 3-4 devices and 2-3 links. The performance of the network is dependent on the capacity of the central hub.
- Easy to connect new nodes or devices. In star topology, new nodes can be added easily without affecting the rest of the network. Similarly, components can also be removed easily.
- Failure of one node or link doesn't affect the rest of the network. At the same time, it's easy to detect the failure and troubleshoot it. Centralized management. It helps in monitoring the network.

Ring Topology

A network topology is set up circularly in such a way that they make a closed loop. This way data travels around the ring in one direction and each device on the ring acts as a repeater to keep the signal strong as it travels. Each device incorporates a receiver for the incoming signal and a transmitter to send the data to the next device in the ring. The network is dependent on the ability of the signal to travel around the ring. When a device sends data, it must travel through each device on the ring until it reaches its destination. Every node is a critical link.^[4] In a ring topology, there is no server computer present; all nodes work as a server and repeat the signal. Sending and receiving of data takes place with the help of TOKEN.

Token passing

The token contains a piece of information which along with data is sent by the source computer. This token then passes to the next node, which checks if the signal is intended for it. If yes, it receives it and passes the empty token into the network, otherwise passes the token along with the data to the next node. This process continues until the signal reaches its intended destination. The nodes with token are the ones only allowed to send data. Other nodes have to wait for an empty token to reach them. This network is usually found in offices, schools, and small buildings. The structure is shown in Figure 30.

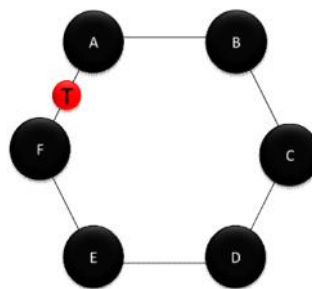


Figure 30 Ring Topology

Advantages of Ring Topology

1. This type of network topology is very organized. Each node gets to send the data when it receives an empty token. This helps to reduce the chances of a collision. Also in a ring topology, all the traffic flows in only one direction at a very high speed.
2. Even when the load on the network increases, its performance is better than that of Bus topology.
3. There is no need for the network server to control the connectivity between workstations.
4. Additional components do not affect the performance of the network.
5. Each computer has equal access to resources.

Disadvantages of Ring Topology

1. The disadvantage of this topology is that if one node stops working, the entire network is affected or stops working.
2. Each packet of data must pass through all the computers between source and destination. This makes it slower than Star topology.
3. If one workstation or port goes down, the entire network gets affected.
4. The network is highly dependent on the wire which connects different components.

Mesh Topology

In a mesh topology, each of the network nodes, computer, and other devices, are interconnected with one another as shown in Figure 31. Every node not only sends its signals but also relays data from other nodes. A true mesh topology is the one where every node is connected to every other node in the network. This type of topology is very expensive as there are many redundant connections, thus it is not mostly used in computer networks. It is commonly used in a wireless network. Flooding or routing technique is used in a mesh topology.

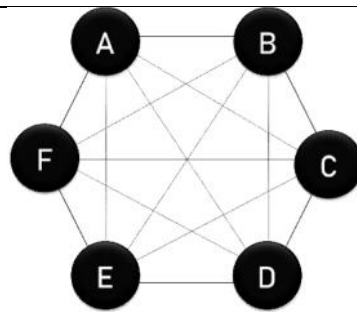


Figure 31 Mesh Topology

Full Mesh Topology

In a full mesh topology, each component is connected to every other component. Even after considering the redundancy factor and cost of this network, its main advantage is that the network traffic can be redirected to other nodes if one of the nodes goes down. Full mesh topology is used only for backbone networks.

Partial Mesh Topology

Here, some of the systems are connected similarly as in mesh topology while the rest of the systems are only connected to 1 or 2 devices. It can be said that in partial mesh, the workstations are 'indirectly' connected to other devices. This one is less costly and also reduces redundancy.

Advantages of Mesh topology

1. Data can be transmitted from different devices simultaneously. This topology can withstand high traffic.
2. Even if one of the components fails there is always an alternative present. So data transfer doesn't get affected.
3. Expansion and modification in topology can be done without disrupting other nodes.

Disadvantages of Mesh Topology

1. There are high chances of redundancy in many of the network connections.
2. The overall cost of this network is way too high as compared to other network topologies.
3. Set-up and maintenance of this topology are very difficult. Even administration of the network is tough.

Tree Topology

Tree Topology integrates the characteristics of Star and Bus Topology. Earlier we saw how in Physical Star network Topology, computers (nodes) are connected by each other through a central hub. And we also saw in Bus Topology, work station devices are connected by the common cable called Bus. In Tree Topology, the number of Star networks are connected using Bus. This main cable seems like the main stem of a tree and other star networks as the branches as shown in Figure 32. It is also called **Expanded Star Topology**. Ethernet protocol is commonly used in this type of topology.

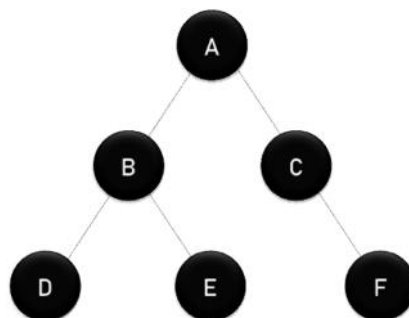


Figure 32 Tree Topology

Advantages of Tree Topology

1. It is an extension of Star and bus Topologies, so in networks where these topologies can't be implemented individually for reasons related to scalability, tree topology is the best alternative.
2. Expansion of Network is possible and easy.
3. Here, we divide the whole network into segments (star networks), which can be easily managed and maintained.
4. Error detection and correction are easy.
5. Each segment is provided with dedicated point-to-point wiring to the central hub.
6. If one segment is damaged, other segments are not affected.

Disadvantages of Tree Topology

1. Because of its basic structure, tree topology, relies heavily on the main bus cable, if it breaks the whole network is crippled.
2. As more and more nodes and segments are added, maintenance becomes difficult.
3. The scalability of the network depends on the type of cable used.

Hybrid Topology

Hybrid, as the name suggests, is a mixture of two different things. Similarly in this type of topology, we integrate two or more different topologies to form a resultant topology that has good points (as well as weaknesses) of all the constituent basic topologies rather than having characteristics of one specific topology. This combination of topologies is done according to the requirements of the organization.



For example:- if there exists a ring topology in one office department while a bus topology in another department, connecting these two will result in Hybrid topology. Remember connecting two similar topologies cannot be termed as Hybrid topology. Star-Ring and Star-Bus networks are the most common examples of the hybrid network as shown in Figure 33.

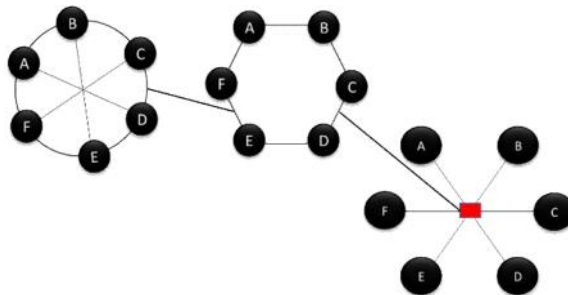


Figure 33 Hybrid Topology

Advantages of Hybrid Network Topology

1. **Reliable:** Unlike other networks, fault detection and troubleshooting are easy in this type of topology. The part in which fault is detected can be isolated from the rest of the network and required corrective measures can be taken, WITHOUT affecting the functioning of the rest of the network.
2. **Scalable:** It's easy to increase the size of the network by adding new components, without disturbing existing architecture.
3. **Flexible:** Hybrid Network can be designed according to the requirements of the organization and by optimizing the available resources. Special care can be given to nodes where traffic is high as well as where chances of fault are high.
4. **Effective:** Hybrid topology is the combination of two or more topologies, so we can design it in such a way that the strengths of constituent topologies are maximized while their weaknesses are neutralized. For example, we saw Ring Topology has good data

reliability (achieved by the use of tokens) and Star topology has high tolerance capability (as each node is not directly connected to the other but through a central device), so these two can be used effectively in hybrid star-ring topology.

Disadvantages of Hybrid Topology

1. **The complexity of Design:** One of the biggest drawbacks of hybrid topology is its design. It's not easy to design this type of architecture and it's a tough job for designers. The configuration and installation process needs to be very efficient.
2. **Costly Hub:** The hubs used to connect two distinct networks are very expensive. These hubs are different from usual hubs as they need to be intelligent enough to work with different architectures and should be functional even if a part of the network is down.
3. **Costly Infrastructure:** As hybrid architectures are usually larger in scale, they require a lot of cables, cooling systems, sophisticated network devices, etc.

Summary

- A network consists of two or more computers that are linked to share resources (such as printers and CD-ROMs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.
- The primary purpose of a computer network is to share resources. The main goal of networking is **Resource sharing**. A second goal is to provide **high reliability** by having alternative sources of supply. Another goal is **saving money**. Another closely related goal is to increase the performance of the system as the workload increases by just adding more processors. With central mainframes, when the system is full, it must be replaced by a larger one, usually at great expense and with even greater disruption to the users. Computer networks provide a powerful communication medium.
- There are two important dimensions for classifying networks — **transmission technology** and **scale**.
- Transmission technology can be classified into two types:
 - o Broadcast networks.
 - o Point-to-point networks
- **Broadcast networks:** These networks have a single communication channel shared by all the machines on the network.
- **Point-to-point** networks consist of many connections between individual pairs of machines. Multiple routes and intermediate machines may exist between a pair of machines, so routing algorithms play an important role here.
- A collection of interconnected networks is called an **internetwork** or just **Internet**. The **Internet** refers to a specific worldwide Internet that is widely used to connect universities, government offices, companies, and private individuals.
- A network topology is the basic design of a computer network. It details how key network components such as nodes and links are interconnected.
- There are three primary types of network topologies that refer to the physical and logical layout of the network cabling. They are a star, ring, and bus topology.

Keywords

Archive: A computer site advertises and stores a large amount of public domain, shareware software and documentation.

Broadcast Networks: They have a single communication channel, which is shared by all the

computers on the network and therefore, any message transmitted by a computer on the network is received by all the computers connected to the channel.

Error Control: The receiving end after the completion of receiving the information must also be capable of dealing with and recognizing corruption.

Local Area Network: A LAN is a form of local (limited distance), shared packet network for computer communications.

Metropolitan Area Network: In MAN, different LANs are connected through a local telephone exchange using one or two cables but not switching elements.

Service Primitives: The primitives enable the service provider to perform some action or report on an action taken by a peer entity.

Wide Area Network: A WAN may be defined as a data communications network that covers a the relatively broad geographic area to connect LANs between different cities with the help of transmission facilities provided by common carriers, such as telephone companies.

Review Questions

Q1: Physical or logical arrangement of a network is

- a) Topology
- b) Routing
- c) Networking
- d) Control

Q2: _____ topology requires a multipoint connection.

- a) Star
- b) Mesh
- c) Ring
- d) Bus

Q3: The types of transmission channel or media used for LAN or WAN are

- a) Twisted Pair Cables
- b) Coaxial Cables
- c) Fibre-Optic Cables and Radio Waves
- d) All of above

Q4: The _____ is the physical path by which a message travels from sender to receiver.

- a) Protocol
- b) Message
- c) Transmission medium
- d) Sender

Q5: A _____ is a set of rules governing data communication between two devices

- a) Protocol
- b) Message
- c) Medium
- d) Sender

Q6: The _____ is the device that sends the message

- a) Protocol
- b) Sender
- c) Network
- d) Medium

Q7: In a _____ connection, two and only two devices are connected by a dedicated link.

- a) Multipoint
- b) Point-to-point
- c) (a) and (b)
- d) None of the above

Q8: An unauthorized user is a network _____ issue.

- a) Performance
- b) Reliability
- c) Security
- d) All the above

Q9: Which topology requires a central controller or a hub?

- a) Mesh
- b) Star
- c) Bus
- d) Ring

Q10: Which type of network would use phone lines?

- a) Wireless
- b) WAN
- c) LAN
- d) WWAN

Q11: Explain five effective characteristics of data communication.

Q12: Write down applications of computer networks.

Q13: How are computer networks classified? Mention some of the important reasons for the classification of computer networks.

Q14: Diagrammatically explain bus topology and ring topology

Q15: Explain the difference between Star, Bus, and Mesh topology.

Q16: Write down the difference between LAN, MAN, and WAN.

Answers

- | | | | | |
|------|------|------|------|-------|
| 1. a | 2. d | 3. d | 4. c | 5. a |
| 6. b | 7. b | 8. c | 9. b | 10. b |

Further/Suggested Readings



- Andrew S. Tanenbaum, *Computer Networks*, Prentice Hall.
- Behrouz A. Forouzan and Sophia Chung Fegan, *Data Communications and Networking*, McGraw-Hill Companies.
- Burton, Bill, *Remote Access for Cisco Networks*, McGraw-Hill Osborne Media, McGraw-Hill Osborne Media.
- Rajneesh Agrawal and Bharat Bhushan Tiwari, *Computer Networks and Internet*, Vikas Publication



- <https://www.geeksforgeeks.org/computer-network-tutorials/>